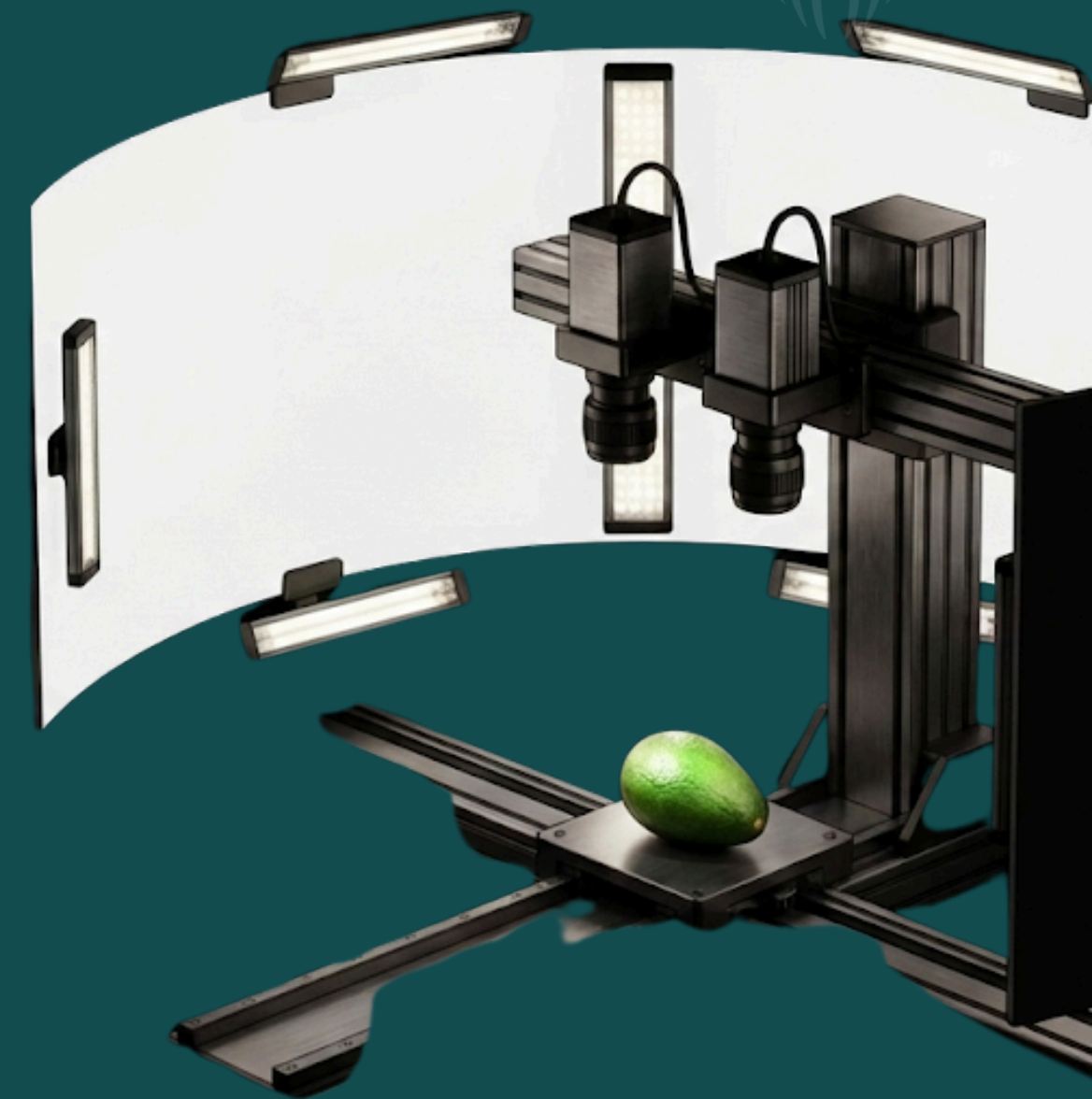


Fruit Ripeness Classification via Hyperspectral Imaging

An Investigation into Non-Destructive Maturity Prediction of Avocados and Kiwis

Pfingsten Julius, **Boscherini** Nicolò,
Piras Ludovico





What is HSI

Hyperspectral Imaging (HSI) is a form of spatial spectroscopy.

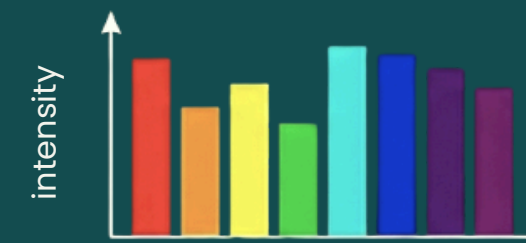
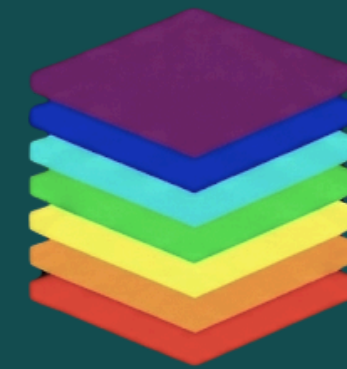
Unlike standard RGB cameras with three color channels, a hyperspectral camera **captures hundreds of channels**, each representing a specific wavelength of light.

This provides a detailed **spectral signature** for every pixel in the image.

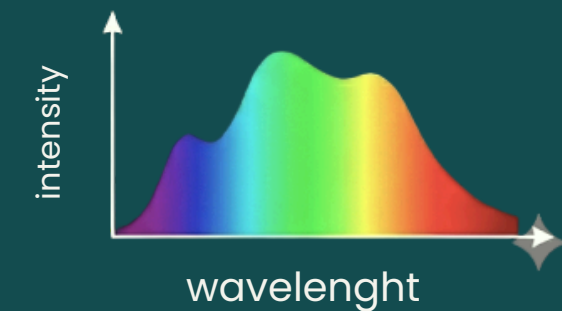
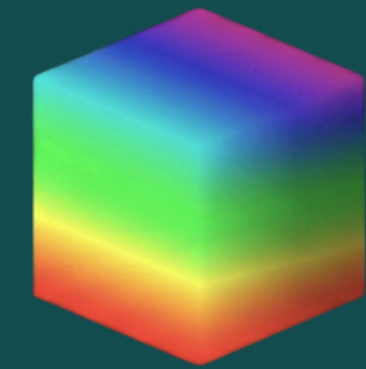
RGB



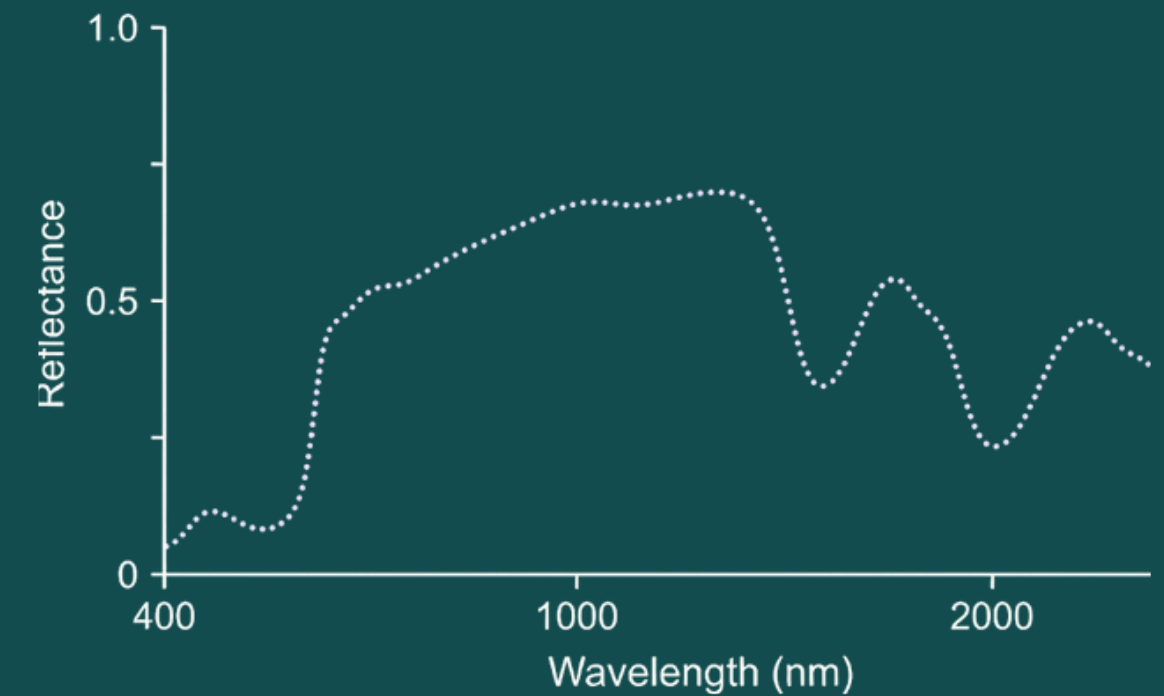
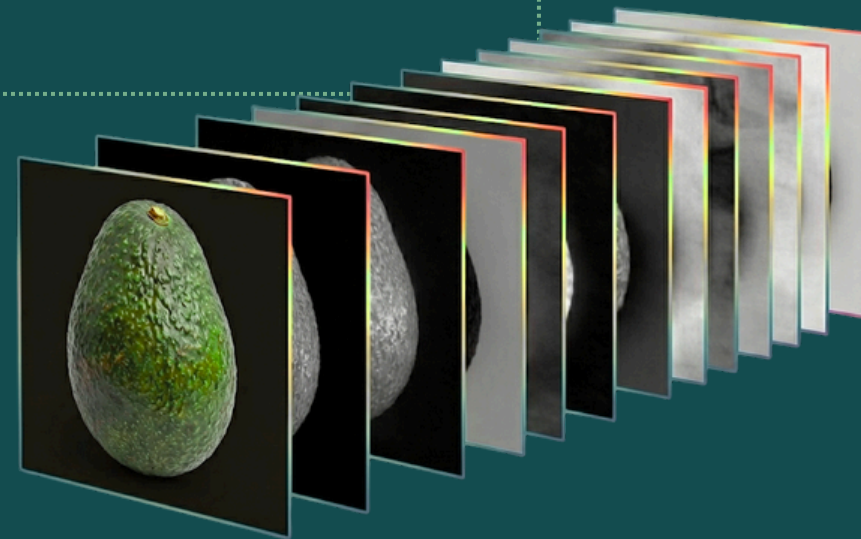
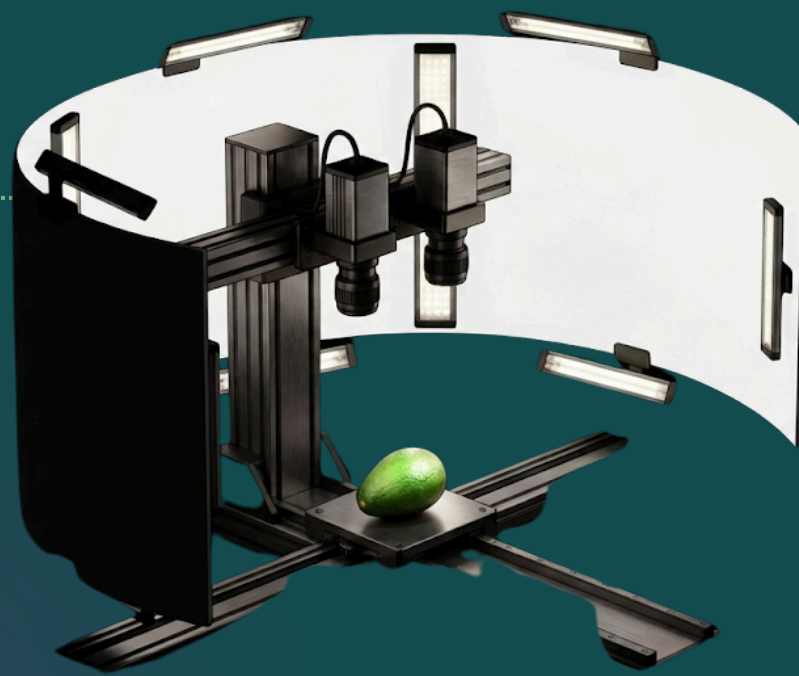
Multispectral



Hyperspectral



Why HSI with Fruit?



The industry relies on **destructive indicator measurements**. This results in:

- **Food waste**
- **Only** random **samples** can be **tested**

HSI allows us to analyze changes in fruit in a **non-destructive** way

Destructive Sampling



Non-Destructive Analysis



Different **wavelengths** reveal different **chemical properties**

The Dataset



The **DeepHS Fruit dataset** (Varga et al., 2021) is the **largest** publicly available hyperspectral **dataset for fruit ripeness prediction**.

Description

The dataset contains a total of 2706 **hyperspectral images** of **five fruit types**, captured by three different cameras.

600 of these **images** are **labeled** by ripeness levels across fruits and cameras.

Data Collection

Images were captured with 3 different cameras:

- Specim FX 10 (400 to 1000 nm)
- Innospec Redeye (920 to 1730 nm)
- Corning microHSI (410 to 900 nm)

Ripeness labels acquired by **destructive** techniques.

Issues

- Very **few labeled images** for one specific camera and fruit
- **Inaccurate splits** into train, test, and val sets **defined by the authors**.
- Inconsistent dataset description

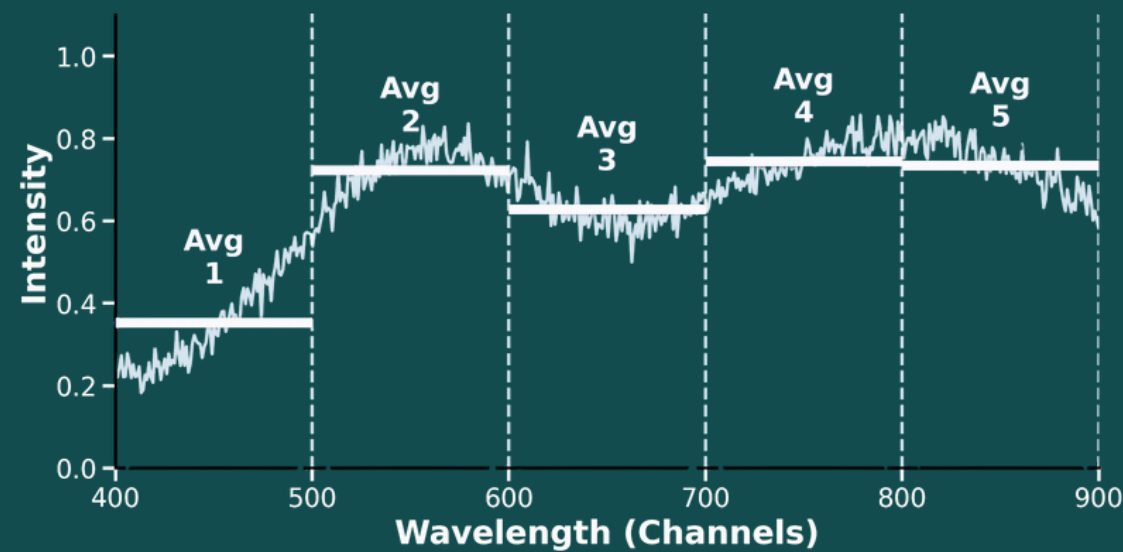
Band Reduction

How did we select the bands?



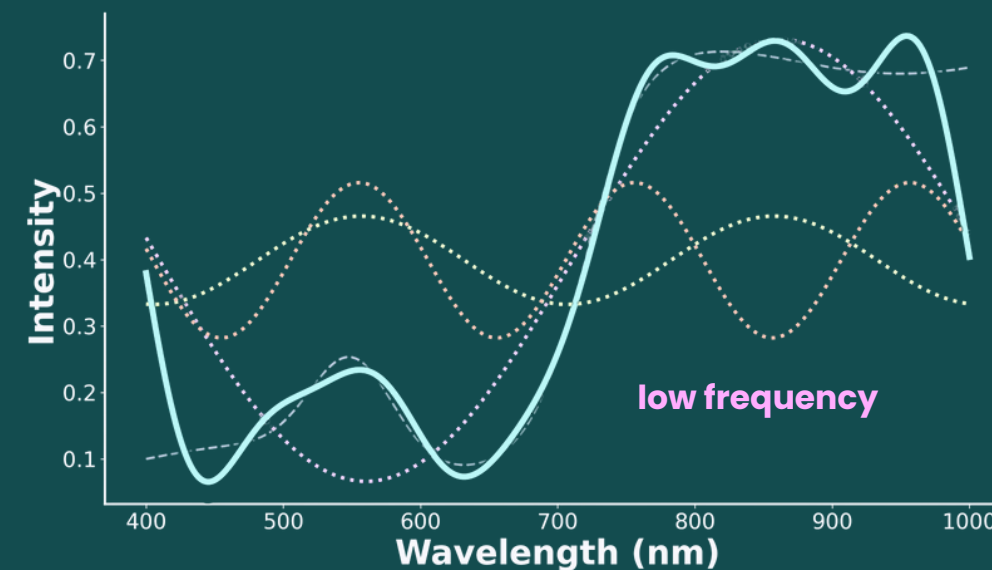
Challenge: HSI images contain hundreds of bands (*channels*). Processing them all is inefficient. We need to reduce them.

Fixed Average



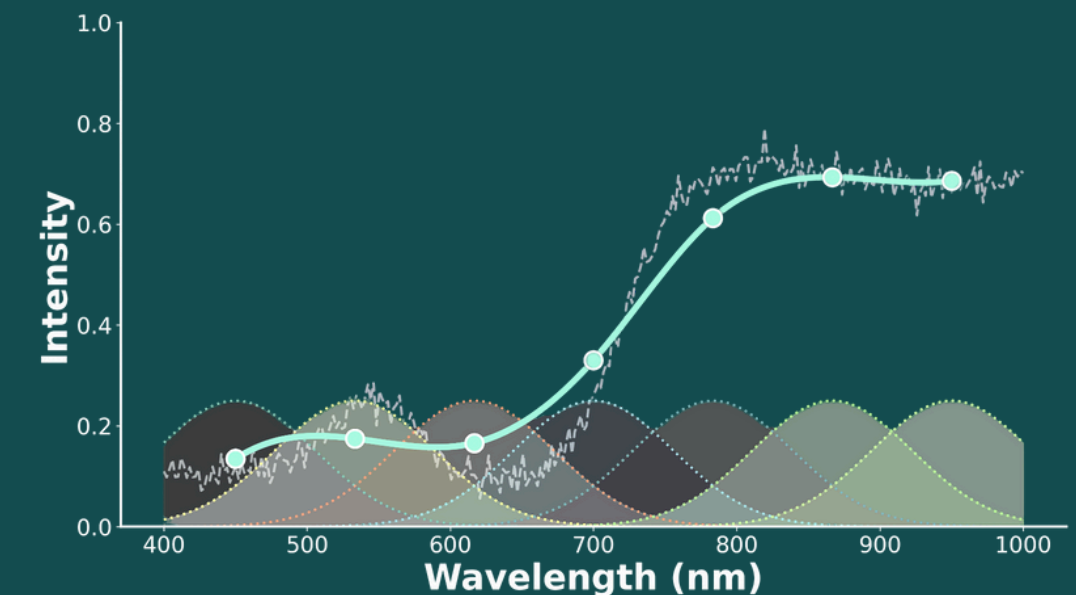
The spectrum is split into k equally sized bins and all the channels falling within these bins are averaged.

Discrete Fourier Transformation



DFT decomposes spectral signatures into frequencies. Retaining the magnitude of the k lowest components acts as a low-pass filter, reducing bands while preserving the overall structure.

Gaussian Weighted-Average

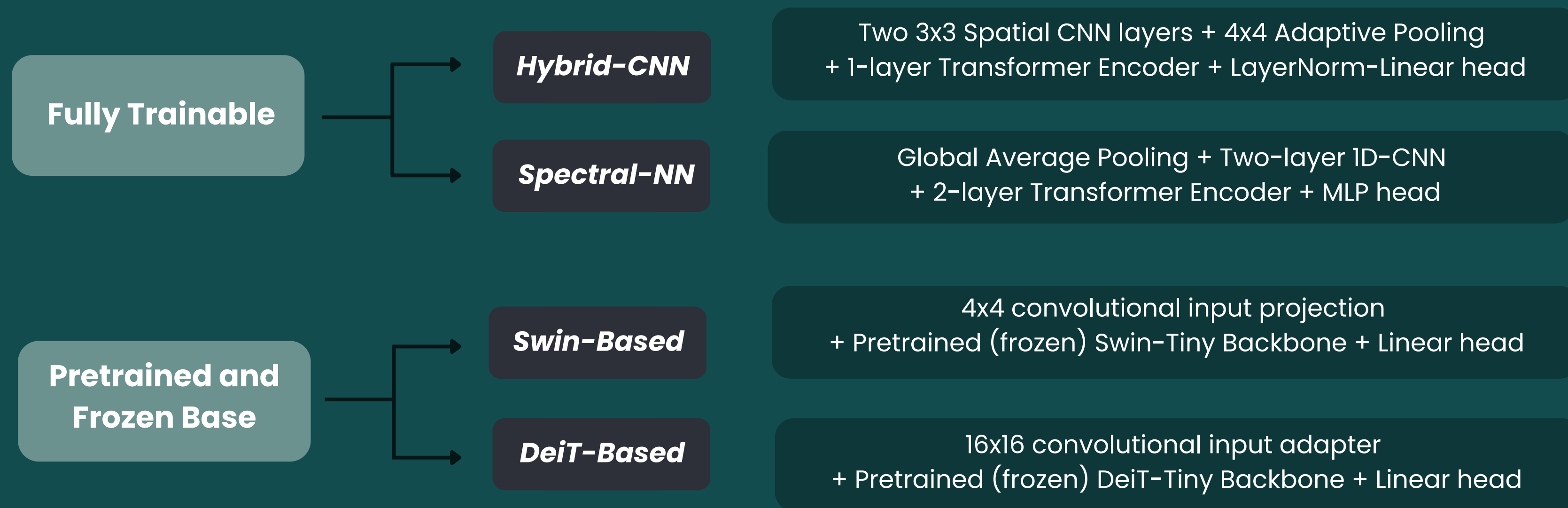


K overlapping Gaussian distributions cover the spectrum. We calculate the normalized, weighted average of channels within each distribution to extract features representing local spectral energy.

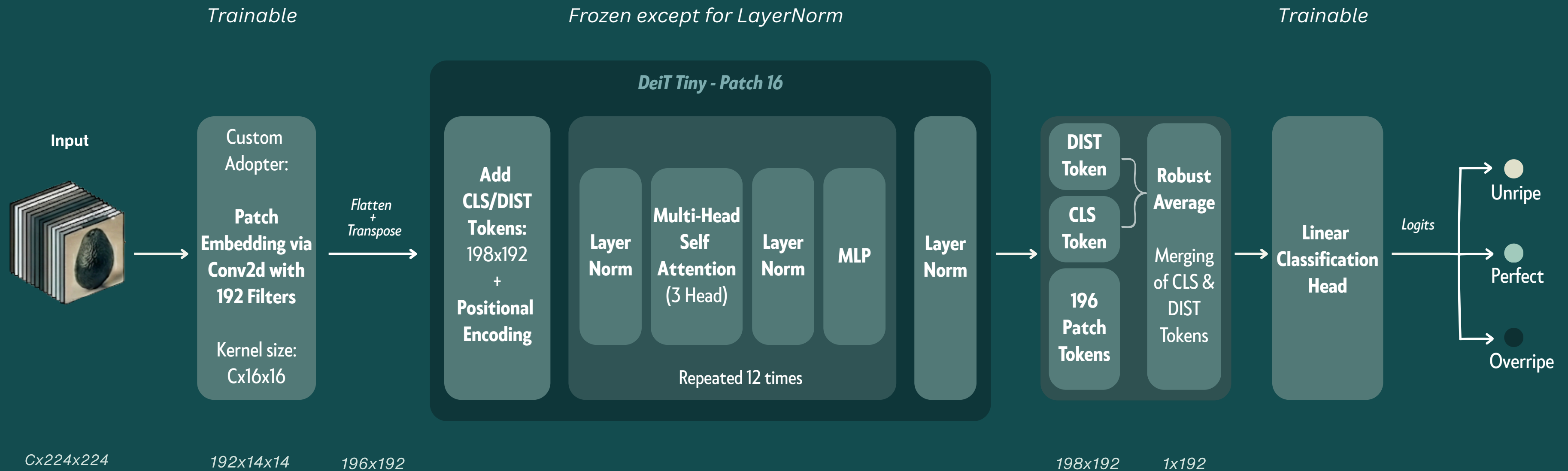
Our Neural Networks



Approach: Explore vision transformers and adapt the attention-based networks to the small HSI dataset



DeiT-Based *Our Best Model*



Note: Swin-Based follows a similar structure, i.e. trainable input adapter feeding a frozen pretrained backbone followed by a linear head

Results



Model	Accuracy (%)		Loss	
	Avocado	Kiwi	Avocado	Kiwi
Hybrid-CNN	71.4 ₁	42.9 ₁	0.85	1.09
Spectral-NN	75 ₂	53.6₃	0.52	0.95
Swin-based	71.4 ₁	42.9 ₁	0.92	1.1
DeiT-based	82.1₁	42.9 ₁	0.53	1.04

Band reduction: 1. DFT 2. Gaussian Weighted-Average 3. Fixed Average

Training performed on Google Colab (NVIDIA T4), 25 epochs, AdamW optimizer (lr = 1e-4), Cross-Entropy loss

Discussion



Performance Analysis

- **Challenge with Kiwis:** Performance was consistently lower for Kiwis compared to Avocados aligning with the baseline findings.
- **The Baseline Gap:** The baseline study (Varga et al., 2021) achieved higher scores using a CNN: 93.3% on Avocado and 66.7% on Kiwis. We did not compare directly because we used a stricter data split to avoid data leakage.
- **The Transformer Constraint:** While we achieved 82.1% accuracy on Avocados, our results confirm that Transformers are data-hungry. The dataset size limited the model's ability to outperform lightweight CNNs.
- **Band reduction** is a key design choice in the HSI setting.

The path forward

- **Why HSI Transformers?** Despite data limitations, the approach is promising. Transformers can capture long-range dependencies between distant spectral bands, linking visible color changes to internal chemical shifts in the NIR range.
- **Scaling Up:** Future work must prioritize expanding the dataset. With sufficient data, Transformers could likely scale better than CNNs.
- **Commercial Viability:** Future systems could reduce data volume by processing only specific "Regions of Interest" (few pixels) or specific band subsets rather than full hypercubes. Companies like OneThird are already deploying avocado scanners.



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Thank You for listening!

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